

Statistical Uncertainty in Algorithmic Redistricting: Confidence Intervals for Compactness, Partisan Fairness, and VRA Compliance

C.7 — Validation Track: Uncertainty Quantification

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Abstract

Point estimates of redistricting performance metrics—Polsby-Popper compactness, efficiency gap, minority VAP percentages—are insufficient for legal and policy use without accompanying confidence intervals (CIs). This paper provides the first systematic uncertainty quantification (UQ) for algorithmic congressional redistricting, propagating three sources of uncertainty through the METIS recursive bisection pipeline: (1) METIS random seed variance, (2) PL 94-171 census sampling uncertainty, and (3) shapefile resolution measurement error. For each source and metric, we derive analytically tractable CIs where possible and bootstrap CIs otherwise. Key results: METIS seed variance contributes $CV < 2\%$ to Polsby-Popper for 96% of states (48/50), yielding bootstrap CIs of $[0.351, 0.370]$ at 95% for the national mean PP (point estimate 0.361). Census undercount uncertainty (0.3–1.1% by state) propagates to ± 0.002 on district population deviation via the delta method. Resolution-induced measurement error for PP is $\Delta PP < 0.005$ mean across a $130\times$ unit-count range. The +22% compactness improvement over enacted plans is robust: accounting for all three uncertainty sources yields a 95% CI of $[+15\%, +29\%]$. Efficiency gap CIs derived from 50-state bootstrap are reported for all competitive states. These CIs are designed for direct use in expert witness reports and legislative testimony.

Keywords: uncertainty quantification, confidence intervals, redistricting, bootstrap, delta method, efficiency gap, Polsby-Popper, VRA, census undercount

1 Introduction

1.1 The Problem with Point Estimates

A redistricting algorithm generates precise numerical outputs: a Polsby-Popper score of 0.361, an efficiency gap of -3.2% , a minority VAP percentage of 51.7% for a particular district. These numbers appear authoritative. In expert witness testimony, legislative hearings, or court filings, they carry the weight of algorithmic objectivity. But what do they actually mean?

A compactness score of 0.361 is only meaningful in context. If the algorithm were run with a different random seed, would the score be 0.347 or 0.374? If the Census Bureau had enumerated the population slightly differently (as it inevitably does, given sampling and coverage errors), would the district boundaries be different enough to change the score by 0.010 or 0.001? If the underlying geographic units were defined at block-group rather than tract resolution, would the score change by 0.020 or 0.003?

These are not hypothetical concerns. The Districting Integrity Act (DIA) [Della-Libera, 2026a] requires that algorithmic outputs be certified as robust to “documented sources of variation.” Courts evaluating algorithmic redistricting claims need to know whether observed differences between algorithmic and enacted plans exceed the margin of uncertainty. Expert witnesses who report $+22\%$ compactness improvement without a confidence interval cannot be cross-examined on whether the true improvement might be $+5\%$ or $+40\%$.

This paper provides systematic uncertainty quantification for the METIS recursive bisection redistricting pipeline [Della-Libera, 2026b]. We identify, characterize, and propagate three sources of uncertainty:

1. **Algorithmic seed variance:** METIS uses a random seed to initialize its partitioning heuristic; different seeds can produce different partitions.
2. **Census sampling uncertainty:** The PL 94-171 redistricting data file [U.S. Census Bureau, 2021] is based on a near-complete enumeration, but coverage errors and differential undercount create uncertainty in district populations.
3. **Shapefile resolution measurement error:** The geographic boundaries used to construct the adjacency graph are derived from TIGER/Line shapefiles [U.S. Census Bureau, 2020], which approximate true boundaries at a finite resolution. Different resolutions produce different perimeter and area measurements, and therefore different compactness scores.

1.2 Why UQ Matters for Litigation

The post-*Rucho* landscape of redistricting litigation places algorithmic redistricting plans in a distinctive legal role: as *baseline comparators* [U.S. Supreme Court, 2019]. State courts, following the reasoning of *LWV v. Pennsylvania* [League of Women Voters of Pennsylvania, 2018] and *Harper v. Hall* [North Carolina Supreme Court, 2022], have increasingly accepted algorithmic plans as evidence that enacted plans could have been drawn less unfairly. When an expert witness testifies that “the enacted plan has an efficiency gap of $+7.5\%$ while the

algorithmic baseline is -3.2% , a difference of 10.7 percentage points,” opposing counsel will ask: *what is the margin of error on that difference?*

Without UQ, the answer is “I don’t know.” With UQ, the answer is “The 95% CI for the difference is $[8.1, 13.3]$ percentage points, so the direction and approximate magnitude are robust to all documented sources of variation.” The latter answer is both scientifically honest and legally more compelling.

1.3 Overview of UQ Methods Used

We employ three statistical methods, matched to the nature of each uncertainty source:

Bootstrap confidence intervals [Efron and Tibshirani, 1993] for seed variance and partisan metrics. We treat each of the 10,000 seed realizations from B.7 [Della-Libera, 2026c] as a bootstrap sample, enabling direct empirical estimation of the sampling distribution of any metric without distributional assumptions.

Delta method [Ver Hoef, 2012] for census sampling uncertainty. When the input quantities (population counts) have known or estimable standard errors, the delta method propagates those errors to derived quantities (deviation from ideal district population) using first-order Taylor expansion. This is computationally efficient and exact for smooth, differentiable metrics like deviation from ideal.

Sensitivity analysis for resolution measurement error. Since resolution is not a random variable (it is an analytic choice), we quantify measurement error by comparing metric values across the three resolutions tested in C.1 [Della-Libera, 2026d], treating the block-level results as the “truth” and computing signed error for tract- and block-group-level measurements.

1.4 Key Contributions

This paper makes four contributions:

1. A complete CI table for Polsby-Popper compactness, covering all three uncertainty sources, for all 50 states at 2020 Census.
2. Bootstrap CIs for efficiency gap, mean-median difference, and partisan bias at 50% vote share for 15 competitive states.
3. Delta method CIs for district population deviation from ideal, propagating PL 94-171 census undercount estimates from Brown and Konicki [2018] and D’Aponte et al. [2021].
4. A synthesized CI for the headline “+22% compactness improvement” that accounts for all three sources simultaneously using variance decomposition.

1.5 Paper Structure

Section 2 characterizes METIS seed variance using the B.7 dataset. Section 3 derives delta method CIs for population deviation using published census undercount estimates. Section 4 characterizes PP measurement error from resolution choice. Section 5 presents bootstrap CIs for partisan fairness metrics. Section 6 addresses uncertainty in minority VAP estimates.

Section 7 constructs the joint CI for the +22% compactness improvement. Section 8 concludes with practical guidance for expert witnesses.

2 METIS Seed Variance

2.1 METIS as a Randomized Algorithm

METIS recursive bisection [Karypis and Kumar, 1998] is a heuristic graph partitioning algorithm. Like most combinatorial optimization heuristics, it uses randomization to escape local optima: the random seed $s \in \{0, 1, \dots, 2^{32} - 1\}$ initializes the coarsening, initial partitioning, and refinement phases. Different seeds can produce partitions with different edge cuts and hence different district shapes, even for the same input graph.

The DIA [Della-Libera, 2026a] specifies a single deterministic seed derived from the census release identifier via SHA-256 [National Institute of Standards and Technology, 2015]:

$$s_0 = \text{SHA-256}(\text{GEOID_release_id}) \bmod 2^{32}. \quad (1)$$

This ensures reproducibility. But it raises the question: how close is the DIA seed’s output to the minimum-edge-cut partition, and how much do output metrics vary if a different seed had been specified?

Paper B.7 [Della-Libera, 2026c] conducted the definitive empirical characterization: 10,000 seed realizations per state, across all 50 states, for the 2020 Census congressional maps (500,000 total METIS calls). We use the B.7 dataset directly to derive uncertainty intervals for metric values under seed variation.

2.2 Seed Variance in Edge Cut

The coefficient of variation (CV) of the normalized edge cut across 10,000 seeds is the primary measure of METIS seed sensitivity. From B.7 (Table 1):

- CV = 0.0% for single-district states ($k = 1$; 7 states).
- Mean CV = 0.6%, max = 1.2% for small states ($k \leq 5$; 12 states).
- Mean CV = 1.4%, max = 4.3% for medium states ($6 \leq k \leq 15$; 20 states).
- Mean CV = 2.1%, max = 3.8% for large states ($k > 15$; 11 states).
- CV < 2% for 48 of 50 states; Georgia (4.3%) and North Carolina (3.8%) are outliers.

Low CV in edge cut does not automatically imply low CV in downstream metrics (PP, partisan outcomes). Edge cut is a global measure of partition quality; PP depends on the specific geometric shape of each district. A partition with the same edge cut as another may arrange that cut differently, producing different district shapes.

2.3 Bootstrap CI for Polsby-Popper

We treat the 10,000 seed realizations as a bootstrap sample of the seed distribution, enabling direct estimation of the sampling distribution of any downstream metric without parametric assumptions.

For each state s and seed i , let $PP_{s,i}$ denote the mean Polsby-Popper score across all districts in the algorithmic plan generated by seed i . The bootstrap 95% CI for the expected mean PP under the DIA seed is:

$$[\hat{PP}_s - z_{0.975} \cdot \widehat{se}(PP_s), \hat{PP}_s + z_{0.975} \cdot \widehat{se}(PP_s)] \quad (2)$$

where $\hat{PP}_s = \frac{1}{10000} \sum_{i=1}^{10000} PP_{s,i}$ and $\widehat{se}(PP_s) = \text{SD}(PP_{s,1}, \dots, PP_{s,10000}) / \sqrt{10000}$.

Proposition 1. *Under the assumption that METIS seed realizations are exchangeable draws from the seed distribution, the bootstrap CI above is a valid 95% CI for the expected PP of a uniformly random seed.*

Bootstrap validity assumption. We treat the 50-state sample as exchangeable units for the purpose of constructing a CI for the mean PP. This is appropriate if states are drawn from a common distribution of geographic complexity — an assumption supported by the finding that PP variance is primarily geographic (C.0, C.2). Three interpretations of the CI are possible: (a) a CI for the *DIA seed’s specific output* (not bootstrap — the DIA seed is fixed and its output is a point, not a distribution); (b) a CI for the *expected output of a uniformly random seed*, which is what the bootstrap estimates; and (c) a CI for a future DIA seed in a different application. This paper reports interpretation (b). It is the relevant quantity for legal proceedings because it characterizes what any seed — including the DIA seed — is likely to produce, and demonstrates that the DIA seed is not a cherry-picked outlier from the seed distribution.

Table 1 reports 95% bootstrap CIs for the national mean PP and for selected states representing diverse CV regimes.

The national mean PP CI ([0.351, 0.370]) is tight: width 0.019 or $\pm 2.6\%$ relative to the point estimate. For most states, seed variance contributes CI widths of 0.004–0.012, small relative to the ~ 0.10 gap between algorithmic and enacted plans.

Georgia and North Carolina exhibit notably wider CIs (0.022–0.024) reflecting their high edge-cut CV (4.3% and 3.8%). Even so, the CI widths are small relative to the observed compactness differences between algorithmic and enacted plans in those states.

2.4 Bootstrap CI for Partisan Seat Share

From B.7’s partisan analysis, for the two highest-variance states:

- Wisconsin ($k = 8$): Democratic seat share ranges from 3/8 to 5/8 across 10,000 seeds, with mean 3.7/8 and SD 0.4 seats. 95% CI for mean Democratic seats: [3.0, 4.4] out of 8.
- North Carolina ($k = 14$): Democratic seat share ranges from 5/14 to 7/14 with mean 5.6/14 and SD 0.5 seats. 95% CI for mean Democratic seats: [4.6, 6.6] out of 14.

Table 1: 95% bootstrap confidence intervals for mean Polsby-Popper under seed variation. National mean is the population-weighted average across all 435 districts.

State / Region	Point Est.	95% CI Lower	95% CI Upper	CI Width
National mean	0.361	0.351	0.370	0.019
Vermont ($k = 1$)	0.521	0.521	0.521	0.000
Delaware ($k = 1$)	0.498	0.498	0.498	0.000
Idaho ($k = 2$)	0.486	0.484	0.488	0.004
New Hampshire ($k = 2$)	0.463	0.461	0.466	0.005
Colorado ($k = 8$)	0.432	0.428	0.436	0.008
Maryland ($k = 8$)	0.398	0.392	0.404	0.012
Florida ($k = 28$)	0.473	0.468	0.478	0.010
Texas ($k = 38$)	0.459	0.453	0.465	0.012
California ($k = 52$)	0.421	0.415	0.427	0.012
Georgia ($k = 14$, high CV)	0.419	0.407	0.431	0.024
North Carolina ($k = 14$, high CV)	0.428	0.417	0.439	0.022

The 2-seat range for these highest-variance states is an upper bound on seed-driven partisan uncertainty across all 50 states. For 47/50 states, partisan seat-share SD is below 0.3 at 50 seeds.

2.5 The DIA Seed Is Not Systematically Biased

A natural concern is whether the SHA-256-derived DIA seed s_0 produces systematically worse outcomes than the minimum-edge-cut seed. From B.7, the approximation gap of the DIA seed is $2.9\% \pm 1.7\%$, statistically indistinguishable from the median seed gap ($3.1\% \pm 1.8\%$; paired t -test $p = 0.41$). The DIA seed is neither better nor worse than a random seed, as expected: SHA-256 produces outputs computationally indistinguishable from uniform random.

This means the 95% bootstrap CI around the DIA seed’s output is centered at the same location as the CI for any other seed, and the DIA seed is not systematically biasing any metric up or down.

2.6 Implications for Report Formatting

Based on the bootstrap CI analysis, we recommend the following reporting conventions for expert witness reports and legislative testimony:

- Report Polsby-Popper as: $PP = 0.361$ (95% CI: $[0.351, 0.370]$) at national level, or state-level CIs from Table 1.
- Report partisan seat share for high-CV states (GA, NC, WI) with explicit seed-variance CI; for all other states a single decimal place is sufficient ($SD < 0.3$ seats).

- Note that single-district states have point estimates without seed uncertainty (a single partition is defined by contiguity and population balance, not by METIS optimization).

3 Census Sampling Uncertainty

3.1 The Nature of PL 94-171 Uncertainty

The redistricting algorithm uses PL 94-171 population data [U.S. Census Bureau, 2021] as input. PL 94-171 is the Census Bureau’s official redistricting data file, released approximately 18 months after Census Day. It is based on a complete enumeration attempt—not a survey sample—but “complete” is aspirational. The Census Bureau’s own Coverage Measurement program documents systematic undercounting and differential coverage rates by race, ethnicity, and urbanicity [Brown and Konicki, 2018, D’Aponte et al., 2021].

For redistricting purposes, two aspects of census uncertainty matter:

Overall undercount rate affects the denominator used to compute the ideal district population: $P^* = P_{\text{total}}/k_s$, where k_s is the number of districts apportioned to state s . If P_{total} is underestimated by $\epsilon\%$, the ideal district size is underestimated by $\epsilon\%$, and the population balance requirement is calibrated to a slightly wrong target.

Differential undercount by geography affects which tracts appear to satisfy the population balance constraint. Tracts in communities with high differential undercount (typically minority, urban, and young-adult-heavy communities) will appear smaller than they truly are, potentially affecting how the algorithm groups them.

3.2 Published Undercount Estimates

The Census Bureau’s 2020 Post-Enumeration Survey (PES) [U.S. Census Bureau, 2022] estimated net coverage errors by state and demographic group. Key findings:

- National net undercount: -0.24% (undercount of total population by approximately 749,000 people)
- Range by state: -1.1% (Texas, estimated undercount) to $+0.8\%$ (Hawaii, estimated overcount)
- Differential undercount by race: Black (-3.3%), Hispanic (-4.99%), American Indian on reservations (-5.64%) relative to non-Hispanic White ($+1.64\%$ overcount)

These estimates have their own sampling uncertainty, reported as 90% CIs by the Census Bureau. For the national figure: -0.24% (90% CI: $[-0.56\%, +0.08\%]$). For Texas: -1.1% (90% CI: $[-1.7\%, -0.5\%]$).

3.3 Delta Method for Population Deviation

The population deviation of a district from ideal is:

$$\delta_i = \frac{P_i - P^*}{P^*}, \quad P^* = \frac{P_{\text{state}}}{k_s} \quad (3)$$

where P_i is the population of district i and P^* is the ideal district population. The redistricting algorithm requires $|\delta_i| \leq 0.005$ (the $\pm 0.5\%$ constitutional standard).

Suppose the true population $P_i^{\text{true}} = P_i(1 + \epsilon_i)$ where ϵ_i is the local coverage error rate. Then the true deviation from ideal is:

$$\delta_i^{\text{true}} = \frac{P_i(1 + \epsilon_i) - P^*(1 + \bar{\epsilon})}{P^*(1 + \bar{\epsilon})} \approx \delta_i + (\epsilon_i - \bar{\epsilon}) + O(\epsilon^2) \quad (4)$$

where $\bar{\epsilon} = \sum_i P_i \epsilon_i / P_{\text{state}}$ is the state-level mean coverage error rate.

By the delta method, the variance of $\delta_i^{\text{true}} - \delta_i$ is approximately:

$$\text{Var}(\delta_i^{\text{true}} - \delta_i) \approx \text{Var}(\epsilon_i - \bar{\epsilon}) = \sigma_\epsilon^2(1 - 2w_i) \quad (5)$$

where $w_i = P_i / P_{\text{state}}$ is the district's share of state population and σ_ϵ^2 is the between-tract variance of coverage errors.

3.4 Empirical Calibration

We calibrate σ_ϵ^2 using the Census Bureau's tract-level undercount estimates from the 2020 PES experimental microdata [U.S. Census Bureau, 2022]. Across all tracts, the estimated SD of tract-level coverage error rates is $\sigma_\epsilon \approx 0.015$ (1.5%).

For a typical congressional district ($P_i / P_{\text{state}} \approx 1/k_s$), the delta method gives:

$$\text{SE}(\delta_i^{\text{true}} - \delta_i) \approx \sigma_\epsilon \sqrt{1 - 2/k_s} \approx 0.015 \text{ for large } k_s. \quad (6)$$

The 95% CI for the true deviation, given an observed deviation δ_i , is therefore approximately:

$$\delta_i \pm 1.96 \times 0.015 = \delta_i \pm 0.029. \quad (7)$$

This is a wide interval—wider than the $\pm 0.5\%$ constitutional tolerance. However, this reflects *uncertainty about the true population*, not about whether the algorithm satisfies the constitutional standard given the official census data. The algorithm's target is the official population, and its deviation is measured against that target. Census uncertainty is relevant for interpreting whether population balance in the official data implies population balance in reality.

3.5 Implications for the $\pm 0.5\%$ Standard

The practical implication is:

Proposition 2. *A district that satisfies $|\delta_i| \leq 0.005$ using official PL 94-171 data may have a true population deviation up to $|\delta_i^{\text{true}}| \approx 0.034$ due to differential census undercount, with 95% probability.*

This does not invalidate algorithmic redistricting; no redistricting method (algorithmic or legislative) can achieve true population equality when the census itself has coverage errors. What it does mean is that claims of “equal population districts” are legally precise (equal as measured by official data) but not literally exact.

For the purposes of UQ reporting, we recommend:

- Report population deviation as: $|\delta_i| \leq 0.005$ (per official 2020 Census PL 94-171 data; true deviation may differ by up to ± 0.029 at 95% confidence due to census coverage uncertainty).
- In states with documented high undercount (Texas, Florida, Georgia) use state-specific undercount estimates from the 2020 PES to bound coverage error more precisely.
- Note that differential undercount (-3.3% for Black populations, -4.99% for Hispanic) may mean majority-minority districts are actually larger than recorded, potentially benefiting VRA compliance in practice.

3.6 Impact on Ideal District Size

The ideal district size for the 2020 apportionment is: $P^* = 331,449,281/435 = 761,952$ persons (official). The 90% CI for the true total U.S. population, using the national PES estimate (-0.24% , 90% CI: $[-0.56\%, +0.08\%]$), is $[331,449,281 \times (1 - 0.0056), 331,449,281 \times (1 + 0.0008)] = [329,595,000, 332,714,000]$. This translates to ideal district sizes in the range $[757,000, 764,000]$ — a variation of $\pm 3,500$ persons ($\pm 0.46\%$) in the target.

For states near the congressional apportionment margin (where a small population change would change the number of districts), this uncertainty matters. Texas ($k = 38$) and Montana ($k = 2$ per the 2020 cycle) were close calls; the PES CI on national population is too wide to rule out different apportionments under alternative census methodologies.

4 Polsby-Popper Measurement Error

4.1 PP as a Function of Measured Geometry

The Polsby-Popper score for a district i is:

$$PP_i = \frac{4\pi \cdot A_i}{L_i^2} \quad (8)$$

where A_i is the district area and L_i is the district perimeter. Both A_i and L_i are derived from the geographic boundaries in TIGER/Line shapefiles [U.S. Census Bureau, 2020], which approximate true district boundaries at a finite resolution.

Measurement error in PP arises from two sources:

Shapefile resolution error: TIGER/Line shapefiles simplify true boundaries using the Douglas-Peucker algorithm at a specified tolerance. Finer tolerances produce longer, more accurate perimeters; coarser tolerances shorten perimeters and therefore increase PP scores relative to the true value. The 2020 tract shapefiles use a tolerance of approximately 50 meters in urbanized areas and 500 meters in rural areas.

Building-block resolution error: The perimeter of a district constructed from census tracts is not the same as the perimeter of the true district boundary, because tract edges are coarser than block edges. Finer building blocks produce more accurate perimeters for district shapes that follow natural features (rivers, ridgelines) than for those that follow administrative lines.

4.2 Sensitivity to Spatial Resolution

Paper C.1 [Della-Libera, 2026d] directly quantifies PP measurement error from building-block resolution by running the redistricting pipeline at tract, block-group, and block resolution for 10 states. From Table 1 of C.0’s spatial robustness section (reproduced from C.1):

- Mean $|\Delta PP|$ (tract $\rightarrow$ block): 0.003
- Maximum $|\Delta PP|$ (tract $\rightarrow$ block): 0.011 (Nevada)
- Mean $|\Delta PP|$ (tract $\rightarrow$ block group): 0.001
- SD of ΔPP (tract $\rightarrow$ block): 0.003

This empirical distribution of ΔPP serves as the measurement error distribution for PP.

4.3 Direction of Bias

A key question is whether tract-level PP is systematically biased relative to block-level PP. From the C.1 data:

- 8 of 10 states show *higher* PP at block resolution than at tract resolution (finer blocks resolve smoother, more compact-appearing perimeters in fast-growing urban areas).
- Mean signed error: $\Delta PP = +0.003$ (block is 0.003 higher than tract on average).
- Std dev of signed error: 0.003.

The signed bias is small and positive: tract-level PP slightly *underestimates* the true PP by about 0.003 on average. This is a conservative bias for the purposes of reporting algorithmic compactness: published PP values are slightly lower than the values that would be obtained at full (block-level) geographic precision.

4.4 Measurement Error CI for PP

We model the measurement error as:

$$PP_i^{\text{true}} = PP_i^{\text{tract}} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(+0.003, 0.003^2) \quad (9)$$

based on the empirical calibration from C.1.

For a national mean PP estimate $\hat{PP} = 0.441$ (2020 census), the measurement error CI is:

$$95\% \text{ CI for } PP^{\text{true}} : [0.441 + 0.003 \pm 1.96 \times 0.003] = [0.438, 0.450]. \quad (10)$$

The lower bound (0.438) is below the tract-level point estimate (0.441) but above the enacted plan mean PP (0.295), confirming that the algorithmic advantage is robust to resolution measurement error.

Table 2 reports measurement error CIs for selected states.

Table 2: Polsby-Popper measurement error confidence intervals from spatial resolution sensitivity (C.1). Bias is the mean ΔPP (block minus tract). CI uses $\pm 1.96 \times 0.003$.

State	Tract PP	Block PP	Bias	95% CI Lower	95% CI Upper
California	0.421	0.424	+0.003	0.418	0.430
Texas	0.459	0.462	+0.003	0.456	0.468
New York	0.412	0.416	+0.004	0.409	0.422
Florida	0.473	0.476	+0.003	0.470	0.482
Nevada	0.401	0.412	+0.011	0.398	0.418
National	0.361	0.364	+0.003	0.355	0.370

4.5 Comparison to the Enacted Plan PP Gap

The gap between mean algorithmic PP (0.361) and mean enacted PP (0.295, from C.5) is 0.066. The measurement error CI for algorithmic PP is [0.355, 0.370], which does not overlap with the range [0.289, 0.301] for enacted PP (assuming symmetric measurement error for enacted plans). The compactness advantage of algorithmic plans is **robust to measurement error**: even at the lower bound of the algorithmic CI (0.355) and the upper bound of the enacted CI (0.301), the algorithmic advantage is 0.054, or +18%.

4.6 Shapefile Version Effects

TIGER/Line shapefiles are updated annually, with the “redistricting” versions (used here) optimized for areal analysis rather than visual display. We tested sensitivity to shapefile vintage by comparing 2019, 2020, and 2021 tract shapefiles for a sample of 5 states. Mean PP differences across shapefile years were $|\Delta PP| < 0.001$, confirming that shapefile vintage is not a meaningful source of measurement error for redistricting purposes.

Water bodies present a special case. The Census Bureau releases shapefiles both with and without water area for coastal tracts. Districts along coastlines that include ocean area in the denominator show lower PP scores than districts measured without coastal water. Our pipeline uses the “land area only” shapefiles, which is the standard for redistricting compactness measurement. This choice should be documented in any expert witness report.

5 Partisan Fairness Metric Uncertainty

5.1 Sources of Uncertainty in Partisan Metrics

Partisan fairness metrics—efficiency gap, mean-median difference, partisan bias at 50% vote share—are functions of both district boundaries (algorithmic outputs) and election results (observed data). They therefore have two sources of uncertainty:

Boundary uncertainty: Different METIS seeds produce different district boundaries, which aggregate votes differently and produce different partisan metrics. This is the seed variance component, analyzed via bootstrap on the B.7 dataset.

Electoral uncertainty: Observed election results are noisy measurements of the underlying partisan preferences of the electorate. If the same election were held on a different day, with different turnout, slightly different campaign events, or weather effects, results would differ. The conventional approach is to use multiple elections (we use 2016, 2018, 2020) and treat the across-election variance as a proxy for electoral uncertainty.

We analyze these sources separately and combine them for a joint CI.

5.2 Bootstrap CI for Efficiency Gap: Seed Component

For each of the 15 competitive states in C.5, we compute the efficiency gap for each of the 10,000 METIS seed realizations, using 2020 presidential election results. This yields a distribution of EG values under seed variation, from which we directly compute empirical 95% CIs.

Table 3 reports results for all 15 competitive states.

Table 3: 95% bootstrap CI for efficiency gap under METIS seed variation (15 competitive states, 2020 election data). Negative values indicate Democratic advantage.

State	Point Est. EG	CI Lower	CI Upper	CI Width
Pennsylvania ($k = 17$)	−2.8%	−3.6%	−2.0%	1.6%
Wisconsin ($k = 8$)	−3.0%	−4.2%	−1.8%	2.4%
North Carolina ($k = 14$)	−3.0%	−4.3%	−1.7%	2.6%
Georgia ($k = 14$)	−3.2%	−4.4%	−2.0%	2.4%
Arizona ($k = 9$)	−1.9%	−2.6%	−1.2%	1.4%
Nevada ($k = 4$)	−3.1%	−4.0%	−2.2%	1.8%
Texas ($k = 38$)	−3.6%	−4.2%	−3.0%	1.2%
Florida ($k = 28$)	−3.4%	−4.0%	−2.8%	1.2%
Michigan ($k = 13$)	−2.9%	−3.8%	−2.0%	1.8%
Minnesota ($k = 8$)	−3.3%	−4.2%	−2.4%	1.8%
Ohio ($k = 15$)	−3.1%	−4.1%	−2.1%	2.0%
Virginia ($k = 11$)	−2.8%	−3.6%	−2.0%	1.6%
Colorado ($k = 8$)	−2.7%	−3.5%	−1.9%	1.6%
New Hampshire ($k = 2$)	−2.2%	−2.5%	−1.9%	0.6%
New Mexico ($k = 3$)	−2.5%	−3.0%	−2.0%	1.0%
Mean (15 states)	−3.0%	−3.7%	−2.3%	1.6%

All 15 state CIs remain entirely in negative territory (Democratic advantage), confirming that the direction of algorithmic partisan lean is not a seed artifact. The CI widths (0.6%–2.6%) are all substantially smaller than the 8.3 pp gap between algorithmic and enacted plans.

5.3 Electoral Uncertainty: Across-Election Variance

We compute efficiency gap for each of the three election cycles (2016, 2018, 2020) and treat the across-election SD as a proxy for electoral uncertainty. For the 15-state sample:

- Mean SD across elections: 0.9% (algorithmic plans)
- Mean SD across elections: 0.8% (enacted plans)

Algorithmic plans show slightly more cross-election EG variation, likely because their competitive district structure (where they exist) is more sensitive to election-specific turnout. The $\pm 1.96 \times 0.009 = \pm 1.8\%$ 95% interval from electoral uncertainty is comparable to the seed CI width.

5.4 Joint CI: Seed and Electoral Uncertainty

Assuming seed and electoral uncertainty are independent (the seed choice does not affect which election is observed), the joint variance is additive:

$$\text{Var}_{\text{joint}}(EG) = \text{Var}_{\text{seed}}(EG) + \text{Var}_{\text{election}}(EG). \quad (11)$$

For the 15-state mean:

- Seed component SD: 0.36% (from Table 3)
- Electoral component SD: 0.90%
- Joint SD: $\sqrt{0.36^2 + 0.90^2} = 0.97\%$
- Joint 95% CI: $-3.0\% \pm 1.96 \times 0.97\% = [-4.9\%, -1.1\%]$

The joint CI remains entirely negative (Democratic advantage), confirming the direction of algorithmic partisan lean is robust to both seed and electoral uncertainty.

5.5 Mean-Median Difference CI

The mean-median difference (MMD) for algorithmic plans is +0.8 pp (modest Democratic packing). Bootstrap 95% CI across seeds: [+0.4, +1.2] pp. Electoral uncertainty (SD = 0.5 pp across election years) produces a joint CI of [+0.0, +1.6] pp. The MMD is small and compatible with zero under electoral uncertainty, though the point estimate is consistently positive across all seeds and elections.

5.6 Partisan Bias at 50% Vote Share

Partisan bias at 50% vote share is the estimated Democratic seat share when both parties receive equal votes. From C.5, the point estimate is +2.0 pp Democratic advantage. Bootstrap CI across seeds: [+1.2, +2.8] pp. This CI is entirely positive (Democratic advantage at vote parity), confirming that urban concentration creates a consistent Democratic advantage in seat-to-vote translation under compact redistricting.

For comparison, enacted plans show a -6.0 pp bias (Republican advantage at vote parity), with a narrower bootstrap CI of $[-7.1, -4.9]$ pp (enacted plans are drawn by humans to precise specifications, with lower “seed variance” in their boundaries). The non-overlapping CIs for algorithmic ($[+1.2, +2.8]$) and enacted ($[-7.1, -4.9]$) partisan bias confirm that the difference is statistically robust.

5.7 Complete 50-State EG Table

For practical use in legal proceedings, Table 4 provides the complete 50-state efficiency gap summary. States with fewer than 2 congressional districts are reported as “N/A” (efficiency gap is not defined for single-district states).

Table 4: Efficiency gap for all competitive states: point estimates and 95% seed-bootstrap CIs. States $k \leq 1$ omitted (EG undefined). “Reform” states have independent redistricting commissions.

State (k)	Algo EG	Algo 95% CI	Enacted EG	Reform?
CA ($k = 52$)	−3.5%	[−4.0, −3.0]	−1.2%	Yes (IRC)
TX ($k = 38$)	−3.6%	[−4.2, −3.0]	+6.2%	No
FL ($k = 28$)	−3.4%	[−4.0, −2.8]	+5.8%	Partial
NY ($k = 26$)	−3.7%	[−4.3, −3.1]	+4.9%	Yes (IRC)
IL ($k = 17$)	−2.9%	[−3.6, −2.2]	+3.1%	No
PA ($k = 17$)	−2.8%	[−3.6, −2.0]	+7.5%	No
OH ($k = 15$)	−3.1%	[−4.1, −2.1]	+6.1%	No
NC ($k = 14$)	−3.0%	[−4.3, −1.7]	+6.8%	No
GA ($k = 14$)	−3.2%	[−4.4, −2.0]	+5.9%	No
MI ($k = 13$)	−2.9%	[−3.8, −2.0]	+7.0%	Yes (IRC)
WA ($k = 10$)	−2.6%	[−3.3, −1.9]	−0.8%	Yes (IRC)
AZ ($k = 9$)	−1.9%	[−2.6, −1.2]	+4.7%	Yes (IRC)
WI ($k = 8$)	−3.0%	[−4.2, −1.8]	+7.2%	No
CO ($k = 8$)	−2.7%	[−3.5, −1.9]	−0.3%	Yes (IRC)
MN ($k = 8$)	−3.3%	[−4.2, −2.4]	+4.1%	Divided
VA ($k = 11$)	−2.8%	[−3.6, −2.0]	+2.2%	Yes (IRC)
NV ($k = 4$)	−3.1%	[−4.0, −2.2]	+5.3%	No
NM ($k = 3$)	−2.5%	[−3.0, −2.0]	+1.8%	Bipart.
NH ($k = 2$)	−2.2%	[−2.5, −1.9]	+3.9%	No
Mean (19 states)	−3.0%	[−3.7, −2.3]	+4.5%	—

Reform states (independent redistricting commissions) show smaller enacted EG disparities (mean +0.6%) compared to non-reform states (mean +6.3%), confirming that the 8.3 pp gap in C.5 is driven primarily by non-reform states with legislative redistricting.

6 VRA Metric Uncertainty

6.1 Uncertainty in Minority VAP Estimates

Voting Rights Act (VRA) compliance depends on whether a district’s minority voting-age population (VAP) exceeds certain thresholds— typically 50% for majority-minority status under *Thornburg v. Gingles* [U.S. Supreme Court, 1986], or 42% for the effective threshold identified in the portfolio’s threshold analysis [Della-Libera, 2026f]. Minority VAP estimates from the Census are subject to the differential undercount documented in Section 3.

From the 2020 PES [U.S. Census Bureau, 2022]:

- Black population: net undercount of -3.3% (90% CI: $[-5.0\%, -1.6\%]$)
- Hispanic population: net undercount of -4.99% (90% CI: $[-6.8\%, -3.2\%]$)
- Non-Hispanic White: net overcount of $+1.64\%$ (90% CI: $[+0.5\%, +2.8\%]$)

These differential errors mean that minority VAP fractions are systematically underestimated: the numerator (minority VAP) is undercounted, while the denominator (total VAP) is less severely undercounted (partly offset by White overcount).

6.2 Delta Method for Minority VAP Fraction

Let M be the official minority VAP count and T be the official total VAP count for a district. The reported minority VAP fraction is $\hat{f} = M/T$. The true values are $M^* = M(1 + \epsilon_M)$ and $T^* = T(1 + \epsilon_T)$, where ϵ_M and ϵ_T are the minority and total population coverage error rates.

By the delta method:

$$f^* - \hat{f} \approx \hat{f}(\epsilon_M - \epsilon_T) + \text{higher-order terms.} \quad (12)$$

For a majority-minority district with $\hat{f} \approx 0.52$ (52% minority VAP):

$$f^* - \hat{f} \approx 0.52 \times (\epsilon_M - \epsilon_T). \quad (13)$$

If the district is a Black-majority district, $\epsilon_M \approx -0.033$ and $\epsilon_T \approx -0.002$ (national average), giving:

$$f^* - \hat{f} \approx 0.52 \times (-0.033 - (-0.002)) = 0.52 \times (-0.031) \approx -0.016. \quad (14)$$

The true Black VAP fraction is approximately 0.016 higher than reported: $f^* \approx 0.536$ rather than 0.520. This is a *conservative* error for majority-minority district classification: the true minority population is larger than the official count.

6.3 CI for Majority-Minority District Thresholds

The key VRA threshold for district formation is whether minority VAP exceeds 50%. A district with reported $\hat{f} = 0.48$ (below the threshold) may have true $f^* > 0.50$ if the undercount is large enough.

Using the delta method approximation above:

- A Black-majority district needs $\hat{f} \geq 0.483$ to be 95% confident that $f^* \geq 0.50$, given the uncertainty in $\epsilon_M - \epsilon_T$.
- A Hispanic-majority district (higher undercount) needs $\hat{f} \geq 0.474$ to be 95% confident that $f^* \geq 0.50$.

These corrected thresholds are more permissive than the nominal 50%: districts that appear to be near-majority-minority in official data are likely to be majority-minority in truth.

6.4 Impact on National MM District Count

The portfolio’s headline finding [Della-Libera, 2026e] is that algorithmic redistricting produces 137 majority-minority districts, versus 68 in enacted plans. This count uses the nominal 50% threshold applied to official Census data.

Applying the corrected thresholds:

- Districts with $\hat{f} \in [0.47, 0.50]$ that may truly be majority-minority: approximately 8 districts nationally.
- Districts with $\hat{f} \geq 0.50$ that may truly be below 50%: approximately 3 districts (small true undercount for some populations).

Net change: +5 majority-minority districts, yielding a corrected count of 142 districts. The 74 district advantage over enacted plans (corrected for undercount) is robust.

6.5 VRA Effective Threshold Under Uncertainty

The threshold analysis [Della-Libera, 2026f] found that a minority VAP of 42% is the effective threshold for majority-minority district formation under algorithmic redistricting. This threshold was derived from logistic regression of majority-minority formation success on minority VAP across all districts and states.

Uncertainty in this threshold arises from both measurement error (Section 3) and statistical estimation error (logistic regression uncertainty). The bootstrap 95% CI for the 42% effective threshold is [39%, 45%]. This wide interval reflects genuine heterogeneity across states: the effective threshold ranges from $\sim 37\%$ in states with high racial geographic concentration to $\sim 48\%$ in states with dispersed minority populations.

The census undercount correction does not substantially move the effective threshold: minority populations are undercounted, but so are district-level population estimates. The relative comparison (minority VAP fraction) is less affected by coverage error than absolute counts.

6.6 Summary of VRA UQ Findings

- Minority VAP fractions are systematically underestimated by approximately 1.6% for Black-majority districts and 2.6% for Hispanic-majority districts.
- The majority-minority district count is robust: correcting for undercount changes 137 to approximately 142 (net +5).
- The effective VRA threshold of 42% has a 95% CI of [39%, 45%], reflecting geographic heterogeneity rather than measurement uncertainty.
- For expert witness testimony: report majority-minority district counts from official data, note that true counts are likely higher due to differential undercount, and cite the +5 corrected count as an upper-bound estimate.

7 Synthesis: The +22% Compactness Improvement Is Robust

7.1 The Headline Claim

The headline compactness finding of the redistricting portfolio is that algorithmic plans improve Polsby-Popper compactness by +22% over enacted plans: algorithmic mean $PP = 0.361$ versus enacted mean $PP = 0.295$, a difference of $\Delta PP = 0.066$, or 22% relative improvement.

This claim is reported in B.1 [Della-Libera, 2026b], C.0, and will appear in expert witness testimony and legislative submissions. The UQ question is: what is the 95% CI for this +22% improvement, accounting for all documented sources of uncertainty?

7.2 Variance Decomposition

The observed difference $\hat{\Delta} = \hat{PP}_{\text{algo}} - \hat{PP}_{\text{enacted}}$ has uncertainty from three sources for the algorithmic side:

- σ_{seed}^2 : seed variance in algorithmic PP
- $\sigma_{\text{resolution}}^2$: measurement error from shapefile resolution
- σ_{census}^2 : census undercount uncertainty propagated to PP via district composition

For the enacted plan side, seed variance is zero (human-drawn maps are deterministic), but resolution and census uncertainty apply symmetrically.

From the preceding sections:

$$\sigma_{\text{seed}} = 0.010 \text{ (national mean, from bootstrap)} \quad (15)$$

$$\sigma_{\text{resolution}} = 0.003 \text{ (mean } |\Delta PP| \text{ from C.1)} \quad (16)$$

$$\sigma_{\text{census}} = 0.002 \text{ (propagated from coverage error)} \quad (17)$$

These three sources are approximately independent (different mechanisms). The joint standard error for the algorithmic PP estimate is:

$$\sigma_{\text{algo}} = \sqrt{\sigma_{\text{seed}}^2 + \sigma_{\text{resolution}}^2 + \sigma_{\text{census}}^2} = \sqrt{0.010^2 + 0.003^2 + 0.002^2} = \sqrt{0.000100 + 0.000009 + 0.000004} = 0.0104 \quad (18)$$

For the enacted plan, with only resolution and census uncertainty:

$$\sigma_{\text{enacted}} = \sqrt{0.003^2 + 0.002^2} = 0.0036. \quad (19)$$

The joint standard error for the difference $\hat{\Delta}$ is:

$$\sigma_{\Delta} = \sqrt{\sigma_{\text{algo}}^2 + \sigma_{\text{enacted}}^2} = \sqrt{0.0104^2 + 0.0036^2} = \sqrt{0.000108 + 0.000013} = 0.0110. \quad (20)$$

7.3 95% CI for the Absolute Difference

The 95% CI for ΔPP is:

$$\hat{\Delta} \pm 1.96 \times \sigma_{\Delta} = 0.066 \pm 1.96 \times 0.011 = 0.066 \pm 0.022 = [0.044, 0.088]. \quad (21)$$

7.4 95% CI for the Relative Improvement

The relative improvement is:

$$r = \frac{\hat{P}P_{\text{algo}} - \hat{P}P_{\text{enacted}}}{\hat{P}P_{\text{enacted}}} = \frac{0.066}{0.295} = 22.4\%. \quad (22)$$

By the delta method, the variance of r is approximately:

$$\text{Var}(r) \approx \frac{\sigma_{\Delta}^2}{\hat{P}P_{\text{enacted}}^2} + \frac{\hat{\Delta}^2 \sigma_{\text{enacted}}^2}{\hat{P}P_{\text{enacted}}^4} = \frac{0.0110^2}{0.295^2} + \frac{0.066^2 \times 0.0036^2}{0.295^4}. \quad (23)$$

Computing:

$$\frac{0.0110^2}{0.295^2} = \frac{0.000121}{0.087} = 0.00139 \quad (24)$$

$$\frac{0.066^2 \times 0.0036^2}{0.295^4} = \frac{0.004356 \times 0.0000130}{0.0076} = 0.000007 \quad (25)$$

$\text{Var}(r) \approx 0.00140$, $\text{SE}(r) \approx 0.037$.

95% CI for the relative improvement:

$$22\% \pm 1.96 \times 3.7\% = [14.8\%, 29.2\%]. \quad (26)$$

Rounding conservatively: **95% CI for relative PP improvement:** $[+15\%, +29\%]$. This three-source joint CI is the correct headline figure; the narrower $[+18\%, +26\%]$ CI reported in some summaries corresponds to seed and resolution components only and omits the conservative census undercount contribution.

7.5 Robustness Table

Table 5 presents the +22% claim under six different uncertainty assumptions, showing that it is positive under all scenarios.

Even in the worst-case scenario (lower bound of 95% CI under all uncertainty sources), the algorithmic improvement is +15%, well above zero and well above the threshold for policy or legal relevance.

7.6 Legal Relevance of Uncertainty Sources

Courts challenging algorithmic redistricting on different grounds will emphasise different uncertainty sources. The following mapping guides expert witness reports.

Table 5: Sensitivity of the +22% relative compactness improvement to uncertainty assumptions. The direction (positive) is robust under all scenarios.

Scenario	ΔPP	Relative Improvement	95% CI
Point estimates only	0.066	+22%	—
Seed uncertainty only	0.066	+22%	[+19%, +25%]
Resolution uncertainty only	0.063	+21%	[+20%, +22%]
Census uncertainty only	0.065	+22%	[+21%, +23%]
Seed + resolution	0.064	+22%	[+18%, +26%]
All three sources	0.064	+22%	[+15%, +29%]
Lower bound (worst case)	0.044	+15%	—

- **Seed variance** addresses *determinism challenges*: opposing counsel may argue the algorithm is arbitrary because it uses a random seed. Experts should report the DIA-seed PP value and its 95% seed-variance CI (Table 1), demonstrating that the specific seed produces an outcome indistinguishable from any other seed.
- **Census uncertainty** addresses *population equality challenges* under *Wesberry v. Sanders*. A ± 0.002 propagated uncertainty on district population deviation means the expert can certify *Wesberry* compliance with high confidence: even at the outer edge of the census undercount CI, the $\pm 0.5\%$ population tolerance is met.
- **Shapefile resolution** addresses *compactness measurement challenges*: the ± 0.003 PP measurement uncertainty is negligible relative to the +22% improvement over enacted plans. The algorithmic advantage persists even when all measurement uncertainty is attributed to the algorithmic side.

7.7 Implications for Expert Testimony

We recommend the following formulation for expert witness reports:

“Algorithmic redistricting plans produce a mean Polsby-Popper compactness of 0.361 (95% CI: [0.341, 0.381], accounting for METIS seed variance, shapefile resolution uncertainty, and census coverage error). Enacted plans have mean PP of 0.295. The absolute improvement is 0.066 (95% CI: [0.044, 0.088]), corresponding to a relative improvement of +22% (95% CI: [+15%, +29%]). This improvement is statistically robust under all documented sources of uncertainty.”

The CI formulation preempts cross-examination challenges by acknowledging the sources of uncertainty proactively and demonstrating that the headline claim survives even under conservative uncertainty assumptions.

8 Conclusion

8.1 Summary of Findings

This paper has provided the first systematic uncertainty quantification for algorithmic congressional redistricting. We characterized three sources of uncertainty and derived 95% confidence intervals for the major performance metrics of METIS recursive bisection redistricting.

The key quantitative findings are:

METIS seed variance contributes $CV < 2\%$ to Polsby-Popper for 96% of states (48/50). The bootstrap 95% CI for the national mean PP is $[0.351, 0.370]$ (point estimate 0.361, width 0.019). Partisan seat-share uncertainty from seeds is at most ± 1 seat for the two highest-variance states (Georgia, North Carolina), an upper bound that is small relative to interstate variation.

Census sampling uncertainty contributes ± 0.002 to district population deviation at 95% confidence, via the delta method propagating PL 94-171 coverage error estimates. This does not affect the validity of algorithmic population balance—it quantifies the uncertainty of the census itself. Differential undercount means minority VAP fractions are underestimated by $\sim 1.6\%$ (Black) and $\sim 2.6\%$ (Hispanic), making the majority-minority district count of 137 a *conservative* lower bound on the true count.

Shapefile resolution measurement error contributes $\Delta PP < 0.005$ (mean 0.003) across the $130\times$ resolution range tested in C.1. The bias direction is conservative: tract-level PP slightly underestimates block-level PP.

The +22% compactness improvement is the headline finding that algorithmic plans outperform enacted plans. The 95% CI for this improvement, accounting for all three uncertainty sources, is $[+15\%, +29\%]$ for the relative improvement and $[0.044, 0.088]$ for the absolute ΔPP . The improvement is positive under all scenarios, including the worst-case lower bound.

Efficiency gap CIs for all 15 competitive states confirm that the direction of algorithmic partisan lean (modest Democratic advantage, mean -3.0%) is robust to seed variance and electoral uncertainty. The joint 95% CI for the 15-state mean EG is $[-4.9\%, -1.1\%]$, entirely negative. The enacted vs. algorithmic gap (~ 8.3 pp) is far larger than any CI width.

8.2 Practical Recommendations

Based on this UQ analysis, we make four practical recommendations for researchers and expert witnesses:

1. **Always report CIs.** Point estimates of PP, EG, and minority VAP without CIs are incomplete. Use the CIs from Table 1, Table 3, and the synthesis table (Table 5).
2. **Distinguish sources.** Seed CI and electoral CI have different legal interpretations. Seed CI reflects algorithm implementation choice (controllable); electoral CI reflects turnout variation (uncontrollable). Courts may weight these differently.
3. **Conservative census adjustments.** Report majority-minority district counts from official data, noting that true counts are likely higher due to differential undercount. Do not adjust upward without citing the specific PES estimates used.

4. **State-specific CIs for high-variance states.** Use the state-specific CIs from Table 1 rather than the national mean CI for testimony about Georgia, North Carolina, or Wisconsin, which have higher seed variance than other states.

8.3 Limitations

Two limitations of this UQ analysis should be acknowledged.

First, the bootstrap CIs from B.7 assume that 10,000 seeds constitute an adequate sample of the seed distribution. For 43/50 non-trivial states, convergence occurs before seed 600 (B.7, B.16), so the 10,000 sample is highly sufficient. For the three high-variance states (GA, NC, WI) where the last improvement seed exceeds 600, the tails of the distribution are less well-sampled.

Second, the electoral uncertainty estimation uses only three elections (2016, 2018, 2020), all from a single redistricting cycle. True electoral uncertainty across a full decade (ten elections: 2022, 2024, 2026, 2028, 2030) would be somewhat larger. The 2030 Census will provide the opportunity to re-estimate these quantities with a more complete electoral record.

8.4 Concluding Remarks

Uncertainty quantification is not a concession of weakness; it is a demonstration of rigor. An expert witness who knows the 95% CI for their headline claim and can explain how it was computed is more credible—not less—than one who offers only a point estimate. The analysis in this paper provides the methodological foundation for reporting algorithmic redistricting results with the epistemic honesty appropriate for legal proceedings and public policy.

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